Evidence Readiness Assessment for Drug Repurposing Using Data Quality, Safety Signals, Network Structure, and Bayesian Uncertainty

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Abstract

Drug repurposing and indication expansion remain important in pharmaceutical development, but the evidence supporting candidate drug–disease hypotheses is often fragmented across clinical trial registries, biomedical literature, safety resources, and terminology systems. Poor data quality can distort repurposing signals by making associations appear stronger, weaker, or more certain than the underlying evidence supports. This study assesses evidence readiness for drug–disease repurposing by integrating preprocessing quality, terminology mapping provenance, literature evidence composition, safety-signal overlap, clinical-trial network structure, and Bayesian uncertainty. ClinicalTrials.gov records were curated into graph-ready drug–condition pairs through terminology standardisation, placebo exclusion, duplicate removal, and retention of unresolved entities for audit. Retrieved biomedical literature was classified as therapeutic, adverse, or irrelevant; adverse-event overlap was used to qualify safety conflict; and graph-derived structural features were integrated with literature-derived priors through Bayesian updating. Pair-level outputs were consolidated into an evidence-quality ledger containing coverage tiers, evidence-readiness scores, posterior uncertainty estimates, and quality categories. The assessment distinguished evidence-ready hypotheses from pairs limited by terminology uncertainty, literature noise, adverse evidence, safety conflict, incomplete structural support, or posterior imprecision. These outputs support transparent prioritisation of candidates for further scientific review. They do not establish clinical efficacy and should not be interpreted as a substitute for biological, preclinical, pharmacovigilance, or clinical validation.

Keywords: Drug repurposing; Evidence readiness; Data quality; Bayesian uncertainty; Clinical trial networks; Evidence quality; Safety signals

1 Introduction

Drug repurposing and indication expansion remain important components of contemporary pharmaceutical development because existing medicines may already have characterised pharmacology, established manufacturing processes, prior human exposure, and partially understood safety profiles [1, 2]. Recent clinical examples illustrate the continuing relevance of this strategy: semaglutide has shown cardiovascular outcome benefit in adults with overweight or obesity and established cardiovascular disease without diabetes [3], and related semaglutide evidence is now extending into broader cardiorenal contexts, including chronic kidney disease populations with or without type 2 diabetes [4]. Tirzepatide has also demonstrated reductions in obstructive sleep apnoea severity among adults with obesity, further illustrating how established metabolic pharmacology may be evaluated in clinically adjacent or mechanistically related indications [5]. These examples do not imply that repurposing is intrinsically low risk or automatically clinically successful; rather, they show that new therapeutic uses for established or related pharmacological agents remain active, clinically relevant, and strategically important within modern drug development [1, 6].
The clinical relevance of repurposing makes the quality of its supporting evidence especially important. In this study, evidence readiness refers to the extent to which the evidence supporting a drug-disease hypothesis is traceable, standardised, relevant, safety-aware, structurally supported, and uncertainty-qualified enough to permit scientific interpretation. Such evidence is rarely contained in a single harmonised source. It is distributed across clinical trial registries, biomedical literature, safety and adverse-event resources, terminology systems such as MeSH, real-world data sources, and network-derived representations of prior drug-disease activity. This fragmentation matters because biomedical data are often heterogeneous, siloed, variably standardised, and difficult to validate across stages of the research lifecycle [7]. Comparable concerns are recognised in regulatory discussions of real-world data and real-world evidence, where source heterogeneity, terminology, metadata, data validity, and fitness-for-purpose influence whether evidence can support decision-making [8, 9]. In drug discovery, the same principle applies to computational analysis: model outputs depend on the quality, consistency, scope, interoperability, and reusability of the underlying data [10].
Model-based prioritisation is insufficient if the evidence base behind a score is not examined. A high-ranking drug–disease pair may reflect literature volume rather than evidential relevance, biased historical trial activity rather than biological plausibility, weak terminology mapping rather than a true disease association, incomplete safety assessment rather than an acceptable benefit–risk profile, or uncertainty that remains hidden behind a single numerical score. These limitations are especially consequential when evidence is summarised without sufficient attention to study design, effect presentation, adverse-event reporting, and benefit-harm interpretation [11]. In repurposing, therefore, the question is not only whether a candidate association receives a high score, but whether the evidence behind that score is reliable enough to interpret.
In this study, we evaluate drug-disease repurposing evidence using data-quality and uncertainty metrics. The assessment examines whether candidate associations are supported by usable source data, reliable terminology mapping, relevant literature,

safety-aware evidence, prior clinical-trial relationships, and quantified uncertainty. These components are consolidated into an evidence-quality ledger that links each drug-disease pair to both its supporting evidence and the limitations affecting its interpretation. The study does not claim clinical validation; instead, it provides an auditable basis for identifying repurposing hypotheses that are evidence-ready for further scientific review and those whose interpretation remains constrained by incomplete, noisy, conflicted, or uncertain evidence.

2 Methods

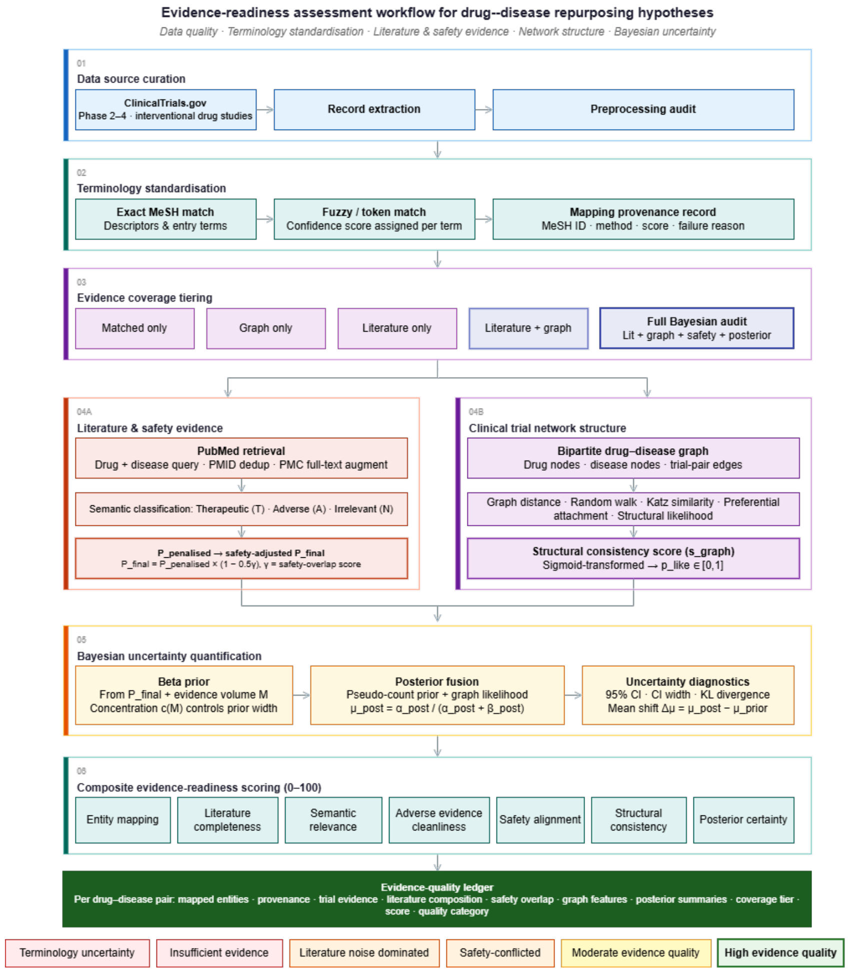

Fig. 1 Evidence-readiness assessment workflow for drug–disease repurposing hypotheses. The workflow links data source curation, terminology standardisation, evidence coverage tiering, literature and safety evidence, clinical-trial network structure, Bayesian uncertainty quantification, composite evidence-readiness scoring, and the final evidence-quality ledger.

2.1 Data source curation and preprocessing readiness

Clinical trial evidence was curated from ClinicalTrials.gov as the primary structured source for historical drug–condition activity [12]. Records were retrieved across predefined therapeutic areas and restricted to interventional drug studies in Phase 2, Phase 3, or Phase 4 with recruitment statuses indicating active, completed, withdrawn, or terminated trial activity. For each eligible record, the trial identifier, therapeutic area, recruitment status, phase, listed conditions, and drug interventions were retained and transformed into trial-derived drug-condition rows for downstream evidence assessment.

Preprocessing readiness was defined as the extent to which the extracted trial data could support an auditable drug-disease evidence graph. The audit captured source volume, drug-condition row generation, terminology matching, unmatched condition terms, unresolved drug terms, placebo-derived records, duplicate drug-condition pairs, and final graph-ready pairs. Placebo interventions were excluded from graph construction, duplicate drug-condition pairs were removed, and unresolved records were retained in the audit trail rather than silently discarded. This allowed source availability, terminology failure, placebo artefacts, duplicate historical activity, and graph-ready evidence to be distinguished explicitly.

The preprocessing audit showed that a large trial-derived evidence base was available, but that only a subset was suitable for graph construction after terminology resolution, placebo exclusion, and de-duplication. Exact extraction, matching, exclusion, and graph-readiness metrics are reported in Table 1.

2.2 Terminology standardisation and mapping provenance

Terminology standardisation was treated as a core evidence-readiness dimension, not merely as a preprocessing operation. Drug and disease names extracted from trial records were cleaned, normalised, and mapped to controlled biomedical terminology using MeSH descriptors and entry terms [13]. This step was required because repurposing evidence often depends on terms that vary across trial records, literature sources, adverse-event resources, and network representations. Without reliable terminology alignment, a candidate drug-disease association may appear unsupported, duplicated, or falsely connected because the underlying entities have not been represented consistently.

Mapping was performed using a staged strategy. Exact matching was applied first to identify terms that aligned directly with MeSH descriptors or recognised entry terms. Where exact matching was not available, fuzzy string matching and token-based similarity were used to identify plausible candidate mappings. Each mapping was assigned a confidence score and a mapping status so that downstream analyses could distinguish high-confidence standardisation from uncertain, partial, or failed mappings. Terms that could not be mapped were not removed from the evidence base; instead, they were preserved with failure reasons to support auditability and to quantify terminology uncertainty.

For each mapped or unresolved term, the terminology audit retained the original term, cleaned term, mapped term, MeSH identifier, mapping method, mapping score, mapping status, and failure reason where applicable. These fields were used to assess the reliability of entity resolution before interpreting literature support, safety overlap, clinical-trial network structure, or posterior uncertainty. In this way, terminology provenance contributed directly to evidence-readiness scoring by identifying which candidate associations were supported by stable entity mappings and which remained limited by unresolved or uncertain nomenclature.

2.3 Evidence coverage tier assignment

Drug-disease pairs were assigned to evidence coverage tiers before interpretation to make the depth of available evidence explicit. Tiering was based on the availability of terminology mapping, usable classified literature, clinical-trial network evidence, safety-overlap assessment, and Bayesian posterior estimation. Pairs were therefore distinguished as matched-only, graph-only, literature-only, literature-and-graph, Bayesian without safety, or full Bayesian audit records.

Coverage tiering was used to prevent incomplete evidence profiles from being interpreted as equivalent to fully assessed pairs. A pair supported only by mapped terminology, literature retrieval, or graph structure was not treated as directly comparable with a pair supported by literature evidence, structural evidence, Bayesian updating, and safety qualification. In the Results, evidence-readiness scores and quality categories are therefore interpreted alongside coverage tier. Full tier counts and the corresponding numbers of unique drugs and diseases are reported in Supplementary Section 2.1 and Table S3.

2.4 Literature retrieval and semantic evidence quality

For each audited drug–disease pair, biomedical literature was retrieved from PubMed using a query that combined the drug term with the target disease term and disease synonyms. Disease terms were expanded to improve recall, while retrieval filters retained records with abstracts and applied the configured date and language restrictions. Retrieved PubMed records were de-duplicated by PMID. Where a PMCID was available, the corresponding PMC full text was queried to augment the record with introduction and conclusion text when these sections could be identified; no text was inferred when PMC sections were unavailable.

Article usability was defined at the record level. A usable record required retrievable bibliographic text, a PMID, an abstract, and a valid semantic classification. Records were classified into one of three mutually exclusive categories: therapeutic, adverse, or irrelevant. Therapeutic articles were those whose available text supported a potential beneficial relationship between the drug and disease; adverse articles were those suggesting harm, worsening, or clinically relevant adverse association; irrelevant articles did not support a meaningful therapeutic or adverse relationship for the specified pair. Classified records were retained with PMID, PMCID where available, title, category, abstract, and any retrieved introduction or conclusion text to support auditability.

Let (T), (A), and (N) denote the numbers of usable articles classified as therapeutic, adverse, and irrelevant, respectively. The raw semantic evidence proportion was defined as:

$$P_{\text{raw}} = \frac{T}{T + A + N}.$$

To penalise adverse evidence within the literature signal, a conservative penalised proportion was also computed:

$$P_{\text{penalised}} = \max\left(\frac{T - 2A}{T + A + N}, 0\right).$$

If no usable classified articles were available, neutral values were retained for downstream uncertainty handling rather than treating absence of literature as evidence for or against a candidate association. These quantities measure the composition and direction of retrieved semantic evidence, not clinical truth. They were used as evidence-quality inputs for the subsequent safety-overlap and Bayesian uncertainty steps and were not interpreted as estimates of therapeutic efficacy.

2.5 Safety-signal overlap assessment

Safety-signal overlap was assessed after semantic literature classification. For each drug–disease pair, post-marketing adverse-event terms were retrieved for the drug and compared with the target disease context. The assessment returned a binary indication of whether a clinically or semantically relevant overlap was present, a set of matching adverse-event terms where applicable, and a safety-overlap score ($\gamma \in [0, 1]$). Higher values of (γ) indicate stronger overlap between reported adverse-event terms and the disease context.

The literature-derived penalised proportion was then adjusted using a bounded safety penalty:

$$P_{\text{final}} = P_{\text{penalised}} \times (1 - 0.5\gamma).$$

When no relevant safety-overlap relation was identified, the penalised proportion was carried forward without additional reduction. The adjustment is conservative: it does not establish that the drug causes harm in the target disease, and it is not a substitute for formal pharmacovigilance or clinical safety assessment. Instead, it reduces evidence readiness when adverse-event signals overlap with the disease context, thereby preventing a literature-supported hypothesis from being interpreted without regard to safety conflict.

2.6 Structural consistency from clinical-trial networks

Structural consistency was assessed using a bipartite drug-disease graph constructed from the graph-ready clinical-trial pairs. Drug entities and disease entities were represented as separate node classes, and an edge was added when a non-placebo drug-disease pair appeared in the processed clinical-trial evidence. The graph therefore represented historical patterns of clinical investigation rather than mechanistic biology or therapeutic efficacy.

For each drug-disease pair, five graph-derived features were computed. Graph distance measured the proximity of a candidate disease to known indications for the same drug using inverse shortest-path information. The random walk score measured

the probability of reaching the disease node from the drug node under a personalised random-walk procedure. Katz similarity quantified connectivity through direct and indirect paths in the bipartite graph. Preferential attachment measured the product of drug and disease degree, capturing whether highly connected entities were more likely to form candidate associations. Structural likelihood combined drug and disease centrality information to summarise whether the pair was supported by the broader topology of historical trial activity.
These features were used to estimate structural consistency as an evidence-readiness dimension. Higher structural consistency indicates that a candidate association is more compatible with observed clinical-trial network patterns.

2.7 Bayesian uncertainty and evidence-readiness scoring

Bayesian updating was used to quantify uncertainty around each audited drug–disease association after semantic literature classification, safety adjustment, and graph-derived structural assessment. The literature-derived and safety-adjusted semantic evidence value, P_{final}, was converted into a Beta prior. Let $M = T + A + N$ denote the number of usable classified articles for the pair. Evidence volume was incorporated through a bounded concentration function:

$$c(M) = c_{\max}\left(1 - \exp\left(-\frac{M}{\tau}\right)\right), \tag{1}$$

where $c_{\max}$ controls the maximum prior concentration and τ controls the rate at which concentration increases with literature volume. The prior distribution was parameterised as:

$$\alpha_{\text{prior}} = 1 + c(M)P_{\text{final}}, \qquad \beta_{\text{prior}} = 1 + c(M)\left(1 - P_{\text{final}}\right). \tag{2}$$

Structural evidence from the clinical-trial network was represented as a likelihood contribution. A weighted graph score, s_{graph}, was computed from the graph-derived features and mapped to a probability scale using:

$$p_{\text{like}} = \frac{1}{1 + \exp(-s_{\text{graph}})}. \tag{3}$$

This likelihood probability was represented as a Beta evidence contribution with fixed likelihood strength λ:

$$\alpha_{\text{like}} = 1 + \lambda p_{\text{like}}, \qquad \beta_{\text{like}} = 1 + \lambda\left(1 - p_{\text{like}}\right). \tag{4}$$

The posterior distribution was obtained by pseudo-count fusion of the literature-derived prior and graph-derived likelihood evidence:

$$\alpha_{\text{post}} = \alpha_{\text{prior}} + (\alpha_{\text{like}} - 1), \qquad \beta_{\text{post}} = \beta_{\text{prior}} + (\beta_{\text{like}} - 1). \tag{5}$$

The posterior mean was calculated as:

$$\mu_{\text{post}} = \frac{\alpha_{\text{post}}}{\alpha_{\text{post}} + \beta_{\text{post}}}. \tag{6}$$

Posterior uncertainty was summarised using posterior variance, the 95% credible interval, credible-interval width, Kullback–Leibler divergence between the posterior and prior Beta distributions, and posterior mean shift:

$$\Delta\mu = \mu_{\text{post}} - \mu_{\text{prior}}. \tag{7}$$

The posterior mean was interpreted as an uncertainty-adjusted evidence-readiness measure, not as a probability of clinical efficacy. Parameter values are reported in Supplementary Section 1.1 and Table S1, and the perturbation design for sensitivity analysis is reported in Supplementary Section 1.2 and Table S2.

2.8 Quality category assignment and evidence-quality ledger

After posterior estimation, each audited drug–disease pair was assigned a composite evidence-readiness score on a 0–100 scale. The score integrated seven bounded quality dimensions: entity mapping, literature completeness, semantic relevance, adverse-evidence cleanliness, safety alignment, structural consistency, and posterior certainty. It was used as a summary measure of audit readiness, while the individual component scores were retained to identify the evidence domain limiting interpretation.

Quality categories were assigned using rule-based thresholds applied to the evidence-readiness score and its limiting factors. These factors included terminology reliability, literature volume, irrelevant retrieval, adverse evidence burden, safety-overlap strength, structural consistency, and posterior uncertainty. The categories distinguished high or moderate evidence quality from insufficient evidence, terminology uncertainty, literature noise dominance, literature conflict, safety-concerning evidence, and safety-conflicted evidence. Full quality-flag definitions are provided in Supplementary Section 2.2 and Table S4.

All pair-level outputs were consolidated into an evidence-quality ledger. For each drug-disease pair, the ledger retained mapped entities, terminology provenance, trial-derived evidence, literature composition, safety-overlap measures, graph-derived structural features, posterior uncertainty summaries, coverage tier, evidence-readiness score, and quality category. The ledger served as the auditable source for the main tables and figures, and a compact pair-level version is provided in Supplementary Section 2.4 and Table S6.

3 Results

3.1 Preprocessing readiness and terminology quality

The preprocessing audit evaluated whether the extracted clinical trial evidence was sufficiently complete, standardised, and traceable to support graph-based drug-disease evidence assessment. Table 1 summarises the main readiness metrics from ClinicalTrials.gov extraction through terminology matching, placebo exclusion, duplicate removal, and final graph construction.

The extraction produced a large trial-derived evidence base, but only a subset was suitable for downstream graph construction after terminology resolution, placebo exclusion, and de-duplication. Of 193,918 generated drug–condition rows, 109,189 were retained after terminology matching. The unmatched condition rate was 11.4%, while the unresolved drug rate was 36.4%, indicating that drug normalisation represented the more substantial terminology bottleneck. After excluding placebo-derived rows and removing duplicate non-placebo drug–condition pairs, 24,156 unique graph-ready pairs remained for structural evidence assessment. The overall mapping success rate was 72.45%, supporting downstream analysis while also showing that terminology uncertainty remained a material evidence-readiness limitation.

Table 1 Preprocessing readiness and terminology-quality audit for trial-derived drug-condition evidence.

Metric	Value
Raw trial records	45,659
Drug-condition rows	193,918
Matched rows after terminology processing	109,189
Unmatched condition rate	0.114
Unresolved drug rate	0.364
Placebo exclusions	580
Duplicate exclusions	84,453
Final graph-ready non-placebo drug-condition pairs	24,156
Mapping success rate	0.7245

Values summarise the clinical-trial extraction, terminology matching, exclusion, and de-duplication audit. Mapping success rate is reported as a proportion. The final graph-ready count represents unique non-placebo drug-condition pairs retained for structural evidence assessment.

3.2 Evidence coverage across drug–disease pairs

Figure 2 shows that evidence coverage was uneven across the audited drug–disease pairs. The coverage tiers distinguish pairs with usable literature evidence, graph-derived structural evidence, Bayesian posterior estimation, and safety-overlap qualification. This distinction is necessary because an evidence-readiness score is only interpretable in relation to the evidence domains available for that pair.

The distribution shows that several pairs had partial rather than complete evidence coverage. Literature-only pairs were semantically assessable but structurally incomplete, while graph-only pairs had network-derived support without usable classified literature. Pairs with Bayesian updating but incomplete safety qualification had stronger multi-domain support than single-domain pairs, but still lacked full safety-aware audit coverage. These distinctions prevent partially assessed hypotheses from being treated as fully evidence-ready.

Figure 2, therefore, supports a central finding of the assessment: coverage tier must be reported alongside evidence-readiness scores, posterior summaries, and quality categories. Full coverage-tier counts, including the corresponding numbers of unique drugs and diseases, are reported in Supplementary Section 2.1 and Table S3.

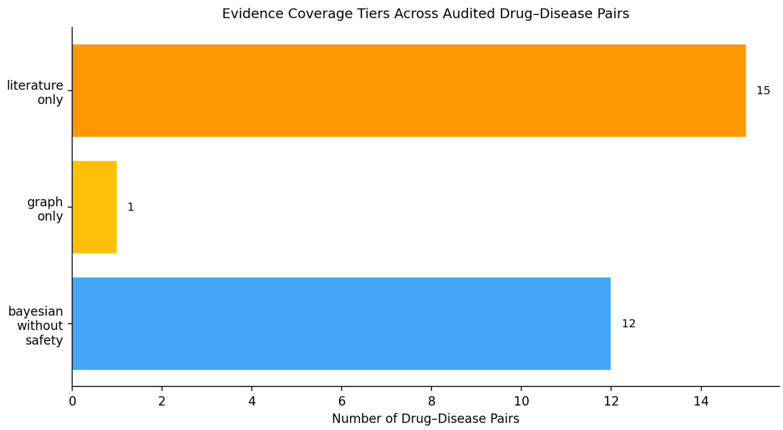

Fig. 2 Evidence coverage tiers across audited drug-disease pairs. The figure distinguishes pairs supported by literature evidence only, graph evidence only, and Bayesian updating without complete safety-overlap coverage. Counts represent the number of audited drug-disease pairs in each coverage tier.

3.3 Case-study evidence-readiness comparison

Figure 3 compares selected drug-disease case studies across the principal evidence-quality dimensions. The heatmap shows that evidence readiness was not determined by a single score, but by the combined behaviour of terminology quality, literature completeness, therapeutic relevance, adverse evidence burden, safety-signal overlap, structural consistency, posterior mean, and overall evidence-readiness score.

Selected case studies were used as retrospective reference anchors to examine whether the assessment separated supportive, contested, conflicted, noisy, safety-concerning, and emerging profiles in plausible directions. These anchors were not treated as clinical validation outcomes. Instead, they provided an interpretability check on whether the evidence-readiness workflow distinguished stronger multi-domain evidence from evidence limited by safety conflict, adverse literature, retrieval noise, terminology uncertainty, or incomplete support.

Table 2 summarises the selected anchors. Positive anchors showed higher posterior means, stronger evidence-readiness scores, and high evidence-quality flags. In contrast, failed, contested, noisy, and conflicted anchors showed lower posterior support or quality flags requiring caution, even when evidence-readiness scores remained relatively high. This distinction is important because a high ERS can reflect auditability and evidence availability, whereas the posterior mean and quality flag indicate the direction and constraint of the evidence. The heatmap complements the tabulated

Table 2 Retrospective reference-anchor comparison across selected case studies.

Drug–disease pair	Reference-anchor role	Posterior mean	ERS	Quality flag
Aspirin–pre-eclampsia	Positive anchor	0.708	87.874	High evidence quality
Colchicine–atherosclerosis	Positive anchor	0.665	86.040	High evidence quality
Dexamethasone–COVID-19	Positive but safety-aware anchor	0.416	79.156	Safety-concerning
Hydroxychloroquine–COVID-19	Failed/conflicted anchor	0.147	77.015	Safety-conflicted evidence
Ivermectin–COVID-19	Contested/high-attention anchor	0.124	73.796	Moderate evidence quality
Azithromycin–albuminuria	Noisy-evidence anchor	0.366	67.069	Literature noise dominated
Levodopa–cardiovascular diseases	Literature-conflicted anchor	0.118	79.371	Literature-conflicted
Metformin–mania	Emerging reviewable hypothesis	0.527	82.752	High evidence quality

ERS denotes the 0–100 evidence-readiness score. Reference-anchor roles were used only for retrospective plausibility checking. The posterior mean is interpreted as an uncertainty-adjusted evidence-readiness signal, not as a probability of clinical efficacy.

anchors by showing that these patterns arise across multiple evidence dimensions rather than from a single summary metric. High-readiness profiles generally showed stronger multi-domain support, whereas safety-concerning, literature-conflicted, noisy, terminology-limited, and structurally incomplete profiles were separated by their limiting evidence domains. Extended pair-level audit details are provided in Supplementary Section 2.4 and Table S6.

3.4 Literature evidence composition and noise

Figure 4 shows the therapeutic, adverse, and irrelevant composition of the retrieved literature across the case-study panel. The distribution confirms that article count alone was not a reliable indicator of evidence quality. Some pairs had substantial retrieval volume with predominantly therapeutic classifications, whereas others contained sizeable adverse or irrelevant components that reduced interpretability.

This distinction was central to the evidence-readiness assessment. Therapeutic classifications indicated directionally supportive semantic evidence, adverse classifications identified potential evidence conflict, and irrelevant classifications quantified retrieval noise. A large retrieved literature set therefore did not automatically strengthen a repurposing hypothesis if the underlying records were mixed, adverse, or poorly aligned with the target drug–disease question.

The literature-composition analysis shows why semantic classification was required before posterior summaries or readiness categories could be interpreted. Article volume was treated as evidence availability, not evidence quality. Extended pair-level literature composition and quality flags are provided in Supplementary Section 2.4 and Table S6.

3.5 Bayesian uncertainty and evidence readiness

Figure 5 shows the relationship between evidence-readiness score and posterior uncertainty, measured by the width of the 95% credible interval. The distribution indicates

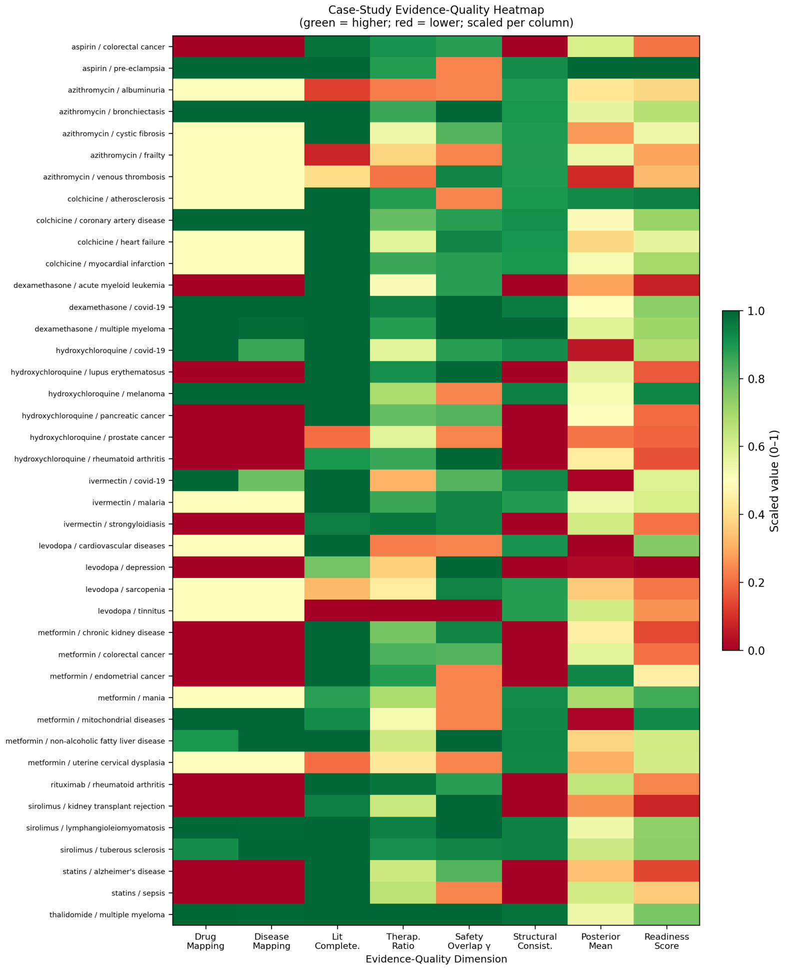

Fig. 3 Case-study evidence-quality heatmap across audited drug–disease pairs. Rows represent drug–disease pairs and columns represent evidence-quality dimensions. Values are scaled within each column from 0 to 1; higher values are shown in green and lower values in red.

that readiness and precision captured related but distinct aspects of evidential support. Pairs with similar readiness scores could have different credible-interval widths and different quality flags, showing that the composite score should not be interpreted without posterior uncertainty.

This distinction was important for safety-concerning, safety-conflicted, and terminology-limited pairs. Some pairs retained moderate or high evidence-readiness scores because several evidence domains were available and auditable, yet their interpretation remained constrained by safety overlap, adverse evidence, terminology uncertainty, or posterior imprecision. Readiness scores therefore summarised audit quality, while credible-interval width qualified how constrained the posterior estimate was.

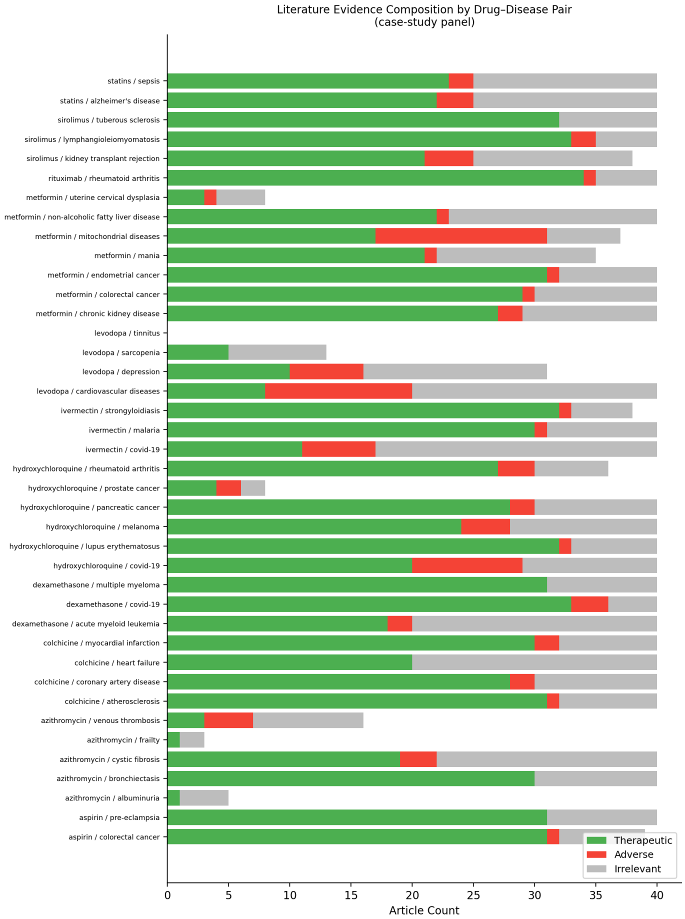

Fig. 4 Literature evidence composition across case-study drug-disease pairs. Bars show retrieved articles classified as therapeutic, adverse, or irrelevant. The figure shows that literature volume and evidence quality diverge when retrieved records contain adverse or irrelevant evidence.

Figure 6 provides an illustrative prior–likelihood–posterior update for azithromycin–albuminuria. The example shows how graph-derived structural evidence can shift the literature-derived prior while preserving uncertainty in the posterior distribution. The posterior mean was therefore interpreted as an uncertainty-adjusted evidence-readiness summary, not as a probability of therapeutic efficacy. Parameter sensitivity results are reported in Supplementary Section 2.3 and Table S5.

3.6 Candidates for further review and low-readiness examples

Table 3 distinguishes evidence-ready review candidates from adjudication-priority pairs. Panel A contains pairs with high evidence-readiness scores, interpretable quality flags, usable literature evidence, and constrained posterior uncertainty. These pairs

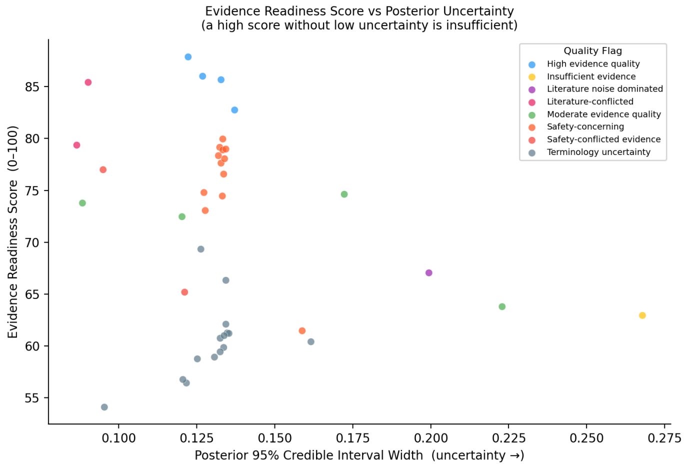

Fig. 5 Evidence-readiness score versus posterior uncertainty across audited drug–disease pairs. Posterior uncertainty is represented by the width of the 95% credible interval. Points are coloured by quality flag, showing that similar readiness scores can correspond to different uncertainty and evidence-quality profiles.

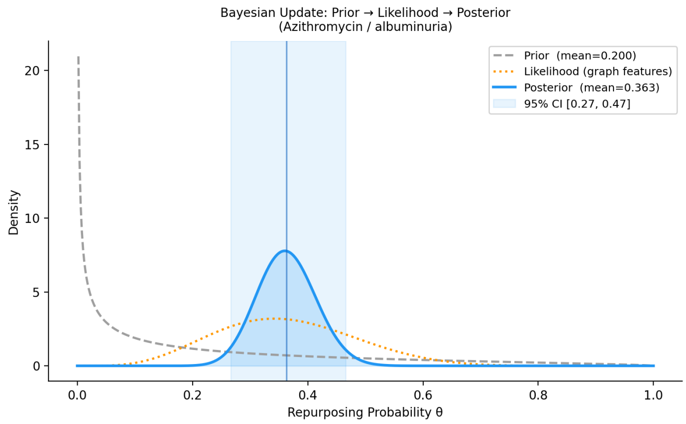

Fig. 6 Illustrative Bayesian update for azithromycin–albuminuria. The plot shows the literature-derived prior, graph-derived likelihood, posterior distribution, posterior mean, and 95% credible interval. The example illustrates how structural evidence can shift the posterior while preserving uncertainty in the final evidence-readiness interpretation.

should be interpreted as hypotheses suitable for further scientific review, not as confirmed therapeutic associations.

Panel B shows why evidence-readiness score and evidence direction must be interpreted together. Some pairs had high audit readiness because the evidence base was available and traceable, but their posterior means, sparse literature, or conflicted evidence profiles indicated a need for expert adjudication before prioritisation.

Table 4 summarises pairs limited by sparse evidence, retrieval noise, adverse literature, safety overlap, or incomplete coverage. These examples show the value of reporting quality flags beside numerical scores. The distinction made by the assessment is not whether a pair appears promising in isolation, but whether its supporting evidence is complete, relevant, safety-aware, structurally supported, and uncertainty-qualified.

Table 3 Evidence-ready and adjudication-priority drug-disease hypotheses for further scientific review.

Drug–disease pair	Coverage tier	Articles	Posterior mean	ERS	Quality flag
Panel A: High-readiness review candidates with supportive evidence profiles					
Aspirin–pre-eclampsia	Full Bayesian	40	0.708	87.874	High evidence quality
Colchicine–atherosclerosis	Bayesian without safety	40	0.665	86.040	High evidence quality
Hydroxychloroquine–melanoma	Full Bayesian	40	0.425	85.705	High evidence quality
Metformin–mania	Bayesian without safety	35	0.527	82.752	High evidence quality
Panel B: Adjudication-priority pairs requiring expert review before prioritisation					
Metformin–mitochondrial diseases	Full Bayesian	37	0.126	85.423	Literature-conflicted
Levodopa–cardiovascular diseases	Bayesian without safety	40	0.118	79.371	Literature-conflicted
Metformin–uterine cervical dysplasia	Bayesian without safety	8	0.298	74.660	Moderate evidence quality; sparse literature
Ivermectin–COVID-19	Full Bayesian	40	0.124	73.796	Moderate evidence quality; contested evidence profile

ERS denotes the 0–100 evidence-readiness score. The posterior mean is interpreted as an uncertainty-adjusted evidence-readiness signal, not as a probability of clinical efficacy. Panel A contains hypotheses with high evidence quality and supportive posterior profiles. Panel B contains pairs retained because they are audit-ready, evidence-rich, structurally assessable, sparse but reviewable, or contested; however, their posterior means, sparse literature, or conflicted evidence profiles indicate a need for expert adjudication before prioritisation. A high ERS in Panel B reflects auditability and review readiness, not therapeutic support.

4 Discussion

This study assessed drug–disease repurposing evidence across terminology provenance, literature composition, safety-signal overlap, clinical-trial network structure, and Bayesian uncertainty. The central finding is that a repurposing hypothesis cannot be interpreted from a posterior score or ranked position alone. Interpretation depends on whether the supporting evidence is traceable, standardised, relevant, safety-aware, structurally supported, and uncertainty-qualified. High evidence readiness was observed where mapped entities were auditable, retrieved literature was aligned with the target hypothesis, adverse and irrelevant evidence were limited, graph-derived support was present, and posterior uncertainty was constrained.

This has direct relevance for pharmaceutical data science and early repurposing review.

Table 4 Low-readiness, conflicted, or safety-concerning examples identified by the evidence-readiness assessment.

Drug–disease pair	Coverage	Articles	Adverse	Noise	ERS	Limiting evidence feature
Levodopa–tinnitus	Graph only	0	0.000	0.000	62.944	Insufficient evidence
Azithromycin–albuminuria	Bayesian without safety	5	0.000	0.800	67.069	Literature noise dominated
Metformin–mitochondrial diseases	Full Bayesian	37	0.378	0.162	85.423	Literature-conflicted
Levodopa–cardiovascular diseases	Bayesian without safety	40	0.300	0.500	79.371	Literature-conflicted
Levodopa–sarcopenia	Bayesian without safety	13	0.000	0.615	61.497	Safety-concerning
Sirolimus–tuberous sclerosis	Full Bayesian	40	0.000	0.200	78.983	Safety-concerning
Dexamethasone–multiple myeloma	Full Bayesian	40	0.000	0.225	78.080	Safety-concerning
Azithromycin–bronchiectasis	Full Bayesian	40	0.000	0.250	76.608	Safety-concerning
Sirolimus–lymphangioleiomyomatosis	Full Bayesian	40	0.050	0.125	78.899	Safety-concerning
Thalidomide–multiple myeloma	Full Bayesian	40	0.075	0.050	79.967	Safety-concerning

Adverse and Noise denote the proportions of usable classified articles labelled adverse and irrelevant, respectively. ERS denotes evidence-readiness score. Safety-concerning examples require safety-signal resolution before prioritisation.

Candidate-ranking workflows can elevate pairs because of literature volume, network proximity, or a favourable posterior mean, but these signals may reflect data availability rather than interpretable evidence. The present assessment separates evidence availability from evidence quality by retaining terminology provenance, semantic literature class, adverse burden, safety overlap, structural consistency, posterior uncertainty, coverage tier, and quality flag. This distinction is essential when candidate lists are used for portfolio screening, trial intelligence, or evidence-gap assessment.

The evidence-quality ledger provides an auditable basis for triage. Rather than returning a single score, it identifies the evidence domain that constrains interpretation. A pair may require terminology repair, literature curation, safety adjudication, structural evidence review, or uncertainty reduction before it can be considered suitable for further scientific review. This structure supports transparent decision-making and reduces the risk that noisy, conflicted, or incomplete evidence is interpreted as therapeutic promise.

The case-study results demonstrate the need for quality categories alongside numerical summaries. High-readiness pairs were separated from literature-conflicted, safety-concerning, terminology-uncertain, noise-dominated, and evidence-sparse profiles.

Some pairs had substantial literature or graph support but remained constrained by adverse evidence or safety overlap. Others had available posterior estimates but weak interpretability because retrieved literature was noisy or misaligned with the target question. These patterns show that evidence readiness measures auditability and interpretability, not clinical efficacy.

Several limitations remain. The assessment used public sources, which may contain incomplete, delayed, duplicated, or inconsistently coded records. Clinical trial registry data may omit relevant design detail, and intervention or condition terms may vary across sponsors and studies. MeSH-based mapping does not resolve every synonym, formulation, combination product, disease subtype, or context-specific indication. LLM-assisted literature classification may introduce error and requires expert-labelled benchmarking. Safety-signal resources are affected by reporting bias, duplicate reports, missing denominators, confounding by indication, and the absence of exposure-adjusted risk estimates. Graph-derived structure reflects historical clinical investigation rather than mechanism, causality, or treatment effect. The posterior mean and credible interval should therefore be interpreted as evidence-readiness summaries, not as estimates of therapeutic efficacy.

Future work should extend the assessment through richer biomedical ontologies, expert adjudication of ambiguous mappings, larger full-audit panels, and external benchmarking against known successful, failed, and abandoned repurposing cases. Mechanistic, pathway, pharmacology, omics, and real-world evidence could be incorporated where data access and governance permit. Such extensions would allow evidence-readiness categories to be tested across broader therapeutic areas and linked more directly to downstream review decisions.

5 Conclusion

This study presents an evidence-readiness assessment of drug-disease repurposing hypotheses using data quality, safety signals, network structure, and Bayesian uncertainty. The assessment does not establish clinical efficacy and does not replace biological, preclinical, or clinical validation. Its value lies in identifying whether the evidence behind a candidate drug-disease pair is complete, standardised, relevant, safety-aware, structurally consistent, and uncertainty-qualified. By making evidence limitations explicit, the assessment supports transparent and scientifically defensible prioritisation of candidates for further review.

Supplementary Information. Supplementary Material is provided with the manuscript and contains the technical audit details supporting the main analysis, including model constants, sensitivity-analysis design, evidence coverage tiers, quality-flag definitions, parameter sensitivity results, and the compact pair-level evidence-readiness audit.

Declarations

Funding: This research received no external funding.
Conflict of interest: The author declares no competing interests.

Ethics approval and consent to participate: Not applicable.
Consent for publication: Not applicable.
Data availability: All trial-derived data used for this study were collected from publicly accessible sources, primarily ClinicalTrials.gov. Processed datasets are available from the corresponding author on reasonable request.
Materials availability: Not applicable.
Code availability: The source code used for data curation, terminology processing, graph construction, feature extraction, Bayesian uncertainty estimation, and evidence-readiness evaluation is available in the project repository at https://github.com/Franosei/Network-Analysis-of-Drug-Repurposing.
Author contribution: F.O. designed the study, developed the methodology, implemented the experiments, and wrote the manuscript.

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